

CMSC828G: Special Topics in Systems for Machine Learning

Sparsity in Training

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DEPARTMENT OF
COMPUTER SCIENCE

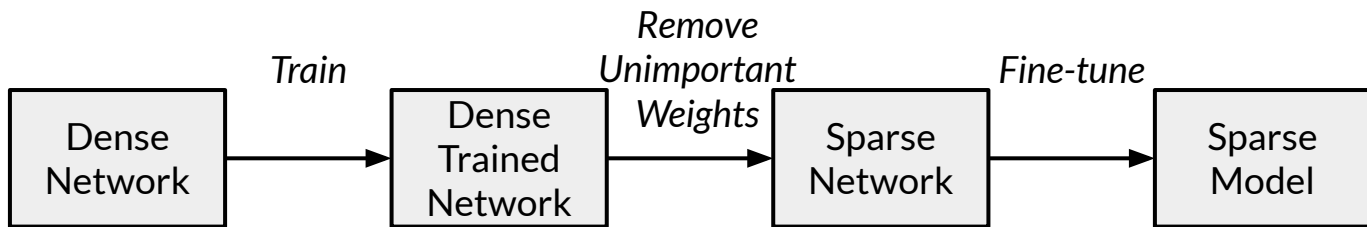
Sparsity: Do We Need All the Computation?

- Dense models perform computation with (almost) all parameters.
- Many computations may be redundant or unnecessary.
- Sparsity attempts to reduce memory, computation, or communication.

Sparsity Comes in Different Forms

- Weight sparsity: some model parameters are inactive / zero (e.g., pruning).
- Attention sparsity: each query attends to only a subset of tokens.
- Activation sparsity: only part of the network executes for an input (e.g., MoE).
- (Gradient sparsity: only selected gradient values are communicated.)

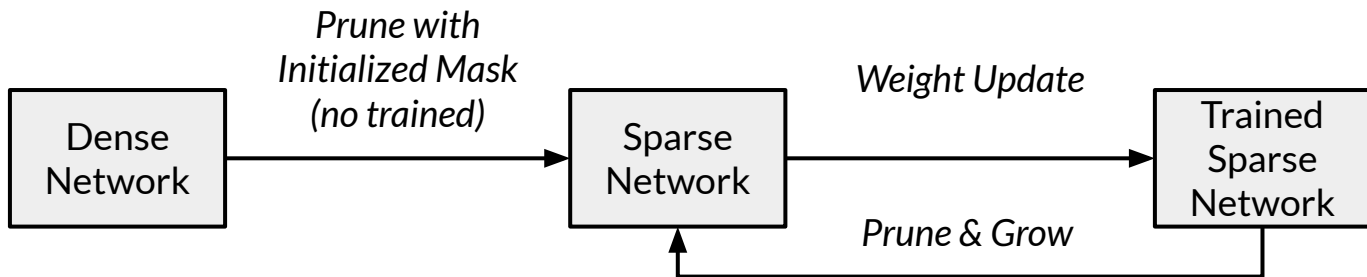
Weight Sparsity: Pruning



$$W_{\text{sparse}} = W \odot M, M \in \{0, 1\}$$

- **Goal:** model compression, inference accelerating while maintaining accuracy.
- **Lottery Ticket Hypothesis:** Dense networks may contain sparse subnetworks that can train successfully from their original initialization^[1].

Weight Sparsity: Sparse Training



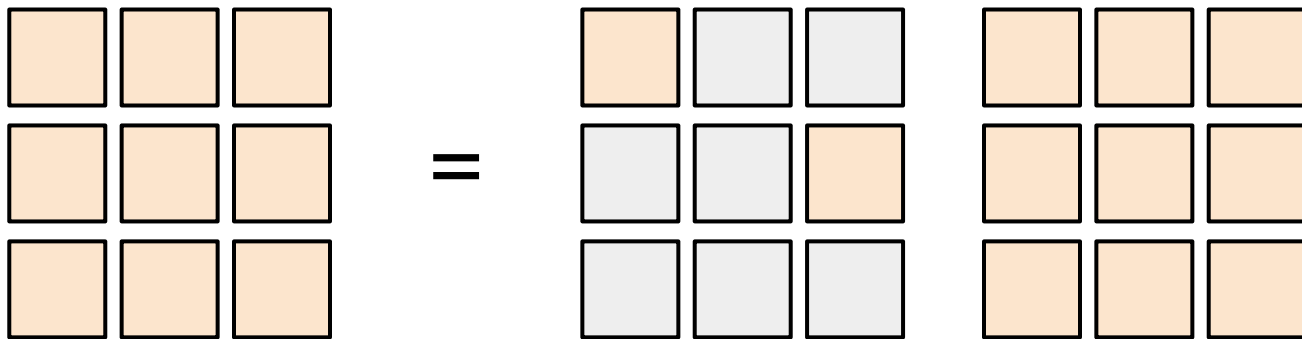
$$W_{\text{sparse}} = W \odot M_t, M_t \in \{0, 1\}, M_{t+1} = \text{Grow}(\text{Prune}(M_t)) \quad ||M_{t+1}||_0 = ||M_t||_0$$

- **Goal:** no dense model training, reducing training cost.
- **Dynamic Sparse Training:** topology changes over time (e.g., RigL^[1]).
- **Static Sparse Training:** sparse mask fixed at the initialization (e.g., SNIP^[2]).

Sparse Computation Primitives: SpMM

- Sparse Matrix-Matrix Multiplication: only nonzeros in **A** do computation.
- CSR / CSC / Block sparse format.

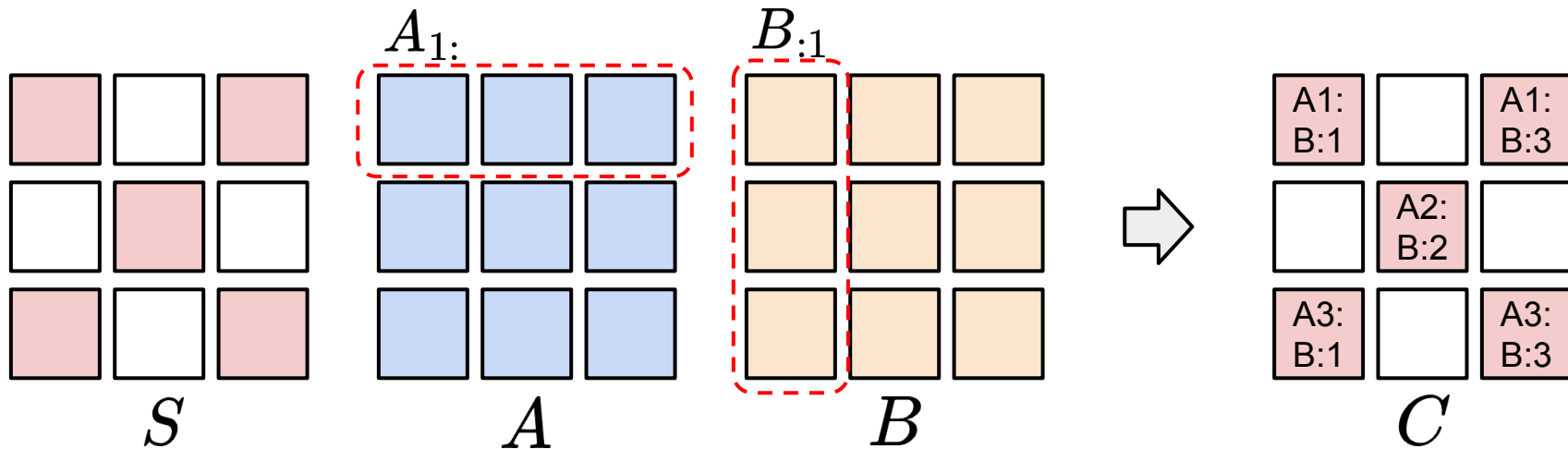
$$C = A_{\text{sparse}} B_{\text{dense}}$$



Sparse Computation Primitives: SDDMM

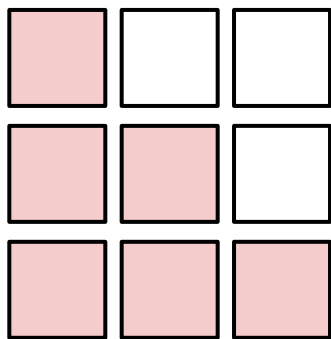
- Sampled Dense-Dense Matrix Multiplication: compute only selected entries of a dense-dense product (rather than dense MM + mask).

$$C = S \odot (AB), C_{ij} = S_{ij} \cdot \langle A_{i:}, B_{:j} \rangle$$

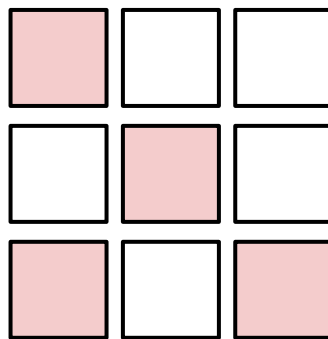


Attention Sparsity

- Full attention: $X = \text{Softmax}(QK^T / \sqrt{d_k})V$
- Sparse attention restricts each token to attend to a small, selected subset of other tokens. $X = \text{Softmax}(M \odot QK^T / \sqrt{d_k})V$
- Real high-performance attention implementations often fuse these operations instead of literally invoking generic SDDMM and SpMM kernels.



Causal
Attention
Mask



Sparse
Attention
Mask

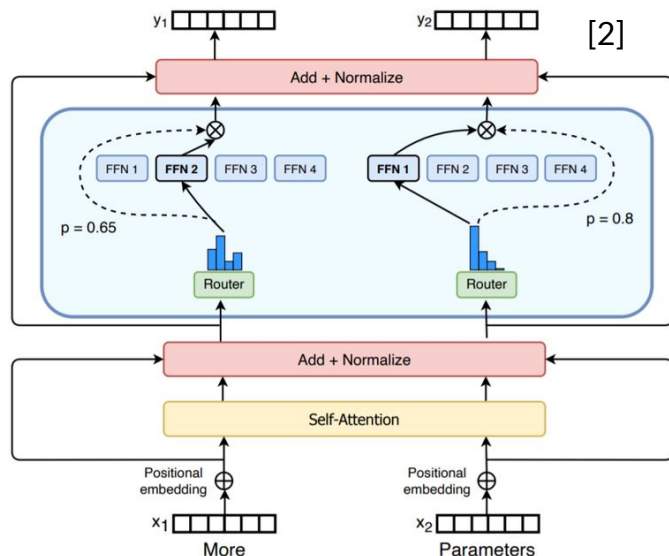
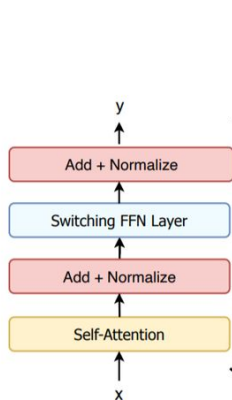
$$X = \text{Softmax}(\underbrace{M \odot QK^T / \sqrt{d_k}}_{\text{SDDMM}}) \underbrace{V}_{\text{SpMM}}$$

The diagram illustrates the fusion of operations in the sparse attention equation. Brackets are used to group the terms: the term $M \odot QK^T / \sqrt{d_k}$ is bracketed and labeled 'SDDMM' (Sparse Double Dot Matrix Multiply), and the term V is bracketed and labeled 'SpMM' (Sparse Matrix-Vector Multiply). The 'Softmax' operation is also indicated by a bracket below the SDDMM term.

Activation Sparsity



- Dense model: every token uses the same parameters.
- Mixture of Experts (MoE)^[1]: every token activates only a subset of experts.
- $N_{\text{expert}}=8$ and $k=2$: 8 experts in total and one token only processes 2 experts.



[1] [ICLR'17] Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer

[2] <https://company.hpc-ai.com/blog/enhanced-moe-parallelism-open-source-moe-model-training-can-be-9-times-more-efficient>

Hardware Support for Sparsity: Sparse Tensor Cores

- Introduced with NVIDIA Ampere GPUs (e.g., A100).
- Exploit 2:4 structured sparsity:
 - In every group of 4 values, 2 must be zero, fixed 50% sparsity pattern.
- Store only the nonzero values + small metadata describing their positions.
- Sparse Tensor Cores **skip multiplications by zero** and operate **directly on the compressed representation**. Can provide up to 2× Tensor Core throughput.

Guest Speaker

- Shaoyi Huang, CS Assistant Professor, Stevens Institute of Technology